THE BEGINNING

Module 3

Slicing 101 — Tier 1: Beginner

1. What is a Slicer?

A slicer is software that converts your 3D model (STL/3MF) into **G-code** — the layer-by-layer instructions your printer actually follows. It "slices" the model into horizontal layers and calculates the exact path the nozzle will travel for each one.

- Popular slicers: Bambu Studio, PrusaSlicer, Cura, OrcaSlicer
- Most are free and support multiple printer brands
- Your printer likely came with a recommended slicer — start there

2. The Slicing Workflow

- Import your model file
- Position and scale it on the virtual bed
- Choose a print profile (or set settings manually)
- Slice — the software calculates the G-code and shows estimated time/material
- Export or send directly to your printer

3. Key Settings Explained Simply

Setting	What it controls
Layer height	Vertical resolution — lower = more detail, slower print
Infill	% of solid material inside the object — affects strength and weight
Print speed	How fast the nozzle moves — faster = quicker but less precise
Supports	Temporary scaffolding printed under overhangs, removed after
Wall count	Number of solid outer layers — affects strength and surface look

4. Reading the Slicer Preview

Before printing, always check the **layer preview** — it shows exactly how the print will build up, including where supports will appear. This is your best chance to catch problems (floating parts, missing supports) before wasting material.

End of Module 3 — Next: Module 4, Print Settings Fundamentals